Trees

Chapter 15

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Trees

- Lists, stacks, and queues are linear in their organization of data
  - Items are one after another.
- In this chapter, we organize data in a nonlinear, hierarchical form
  - Item can have more than one immediate successor

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Terminology

- Trees are hierarchical in nature
  - Means a parent-child relationship between nodes

A is the parent of B and C

B and C are siblings

C, D, E and F are leaves of tree (a)

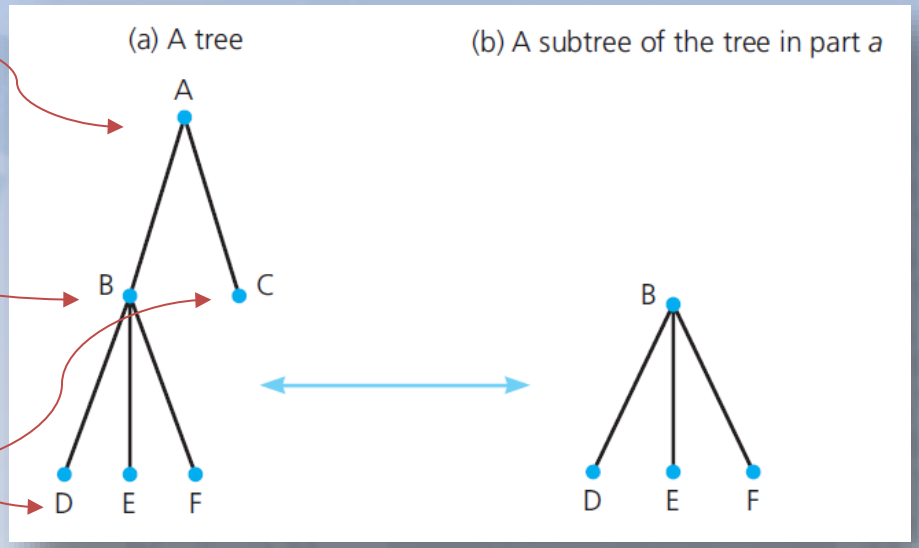

FIGURE 15-1 A tree and one of its subtrees

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Terminology

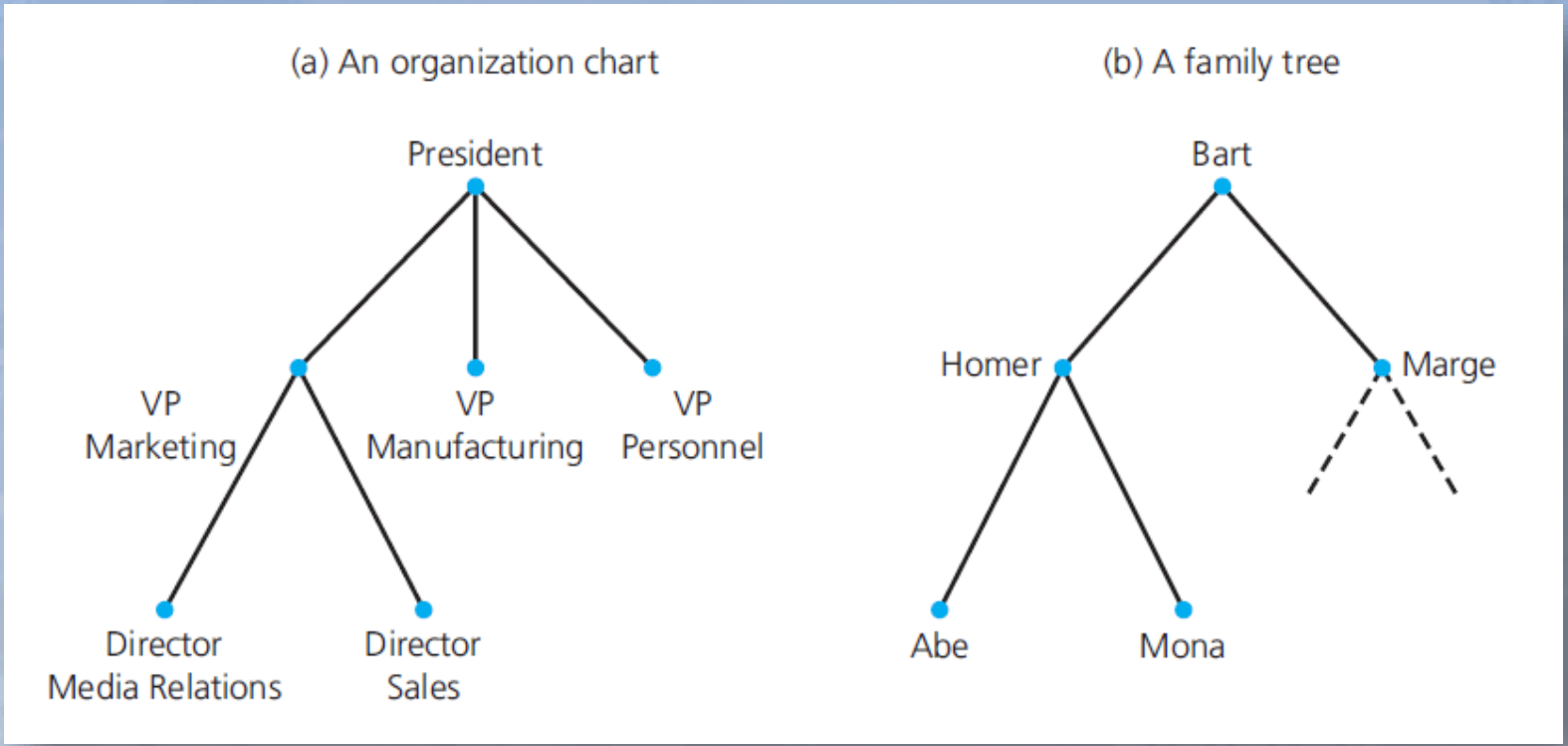

FIGURE 15-2

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Kinds of Trees

- General tree
  - Set *T* of one or more nodes (not empty)
  - *T* is partitioned into disjoint subsets
- An n-ary tree
  - A single node r, the root
  - n possibly empty sets that are n-ary subtrees of r
  - Each node can have no more than n children
- Binary tree
  - Set of *T* nodes – either empty or partitioned into disjoint subsets
  - Single node *r*, the root
  - Two (possibly empty) sets – left and right subtrees

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Kinds of Trees

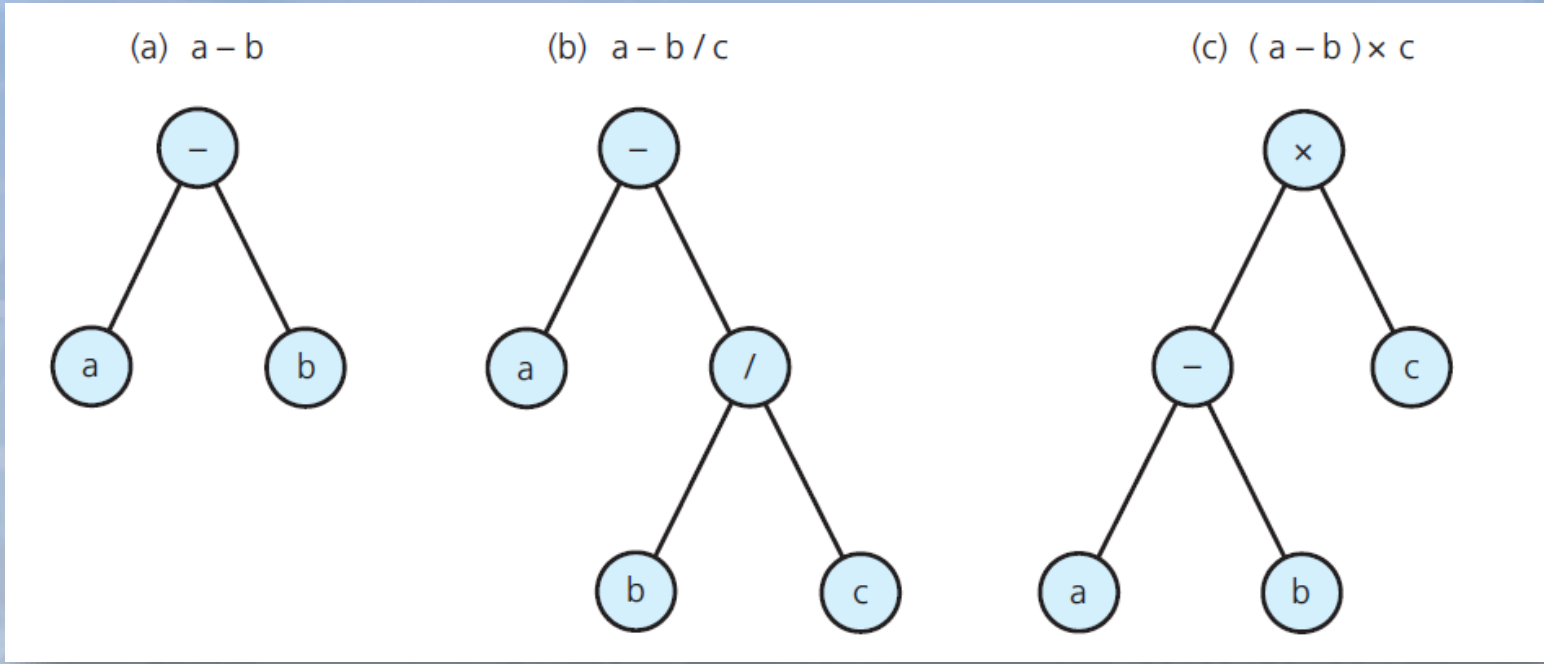

FIGURE 15-3 Binary trees that represent algebraic expressions

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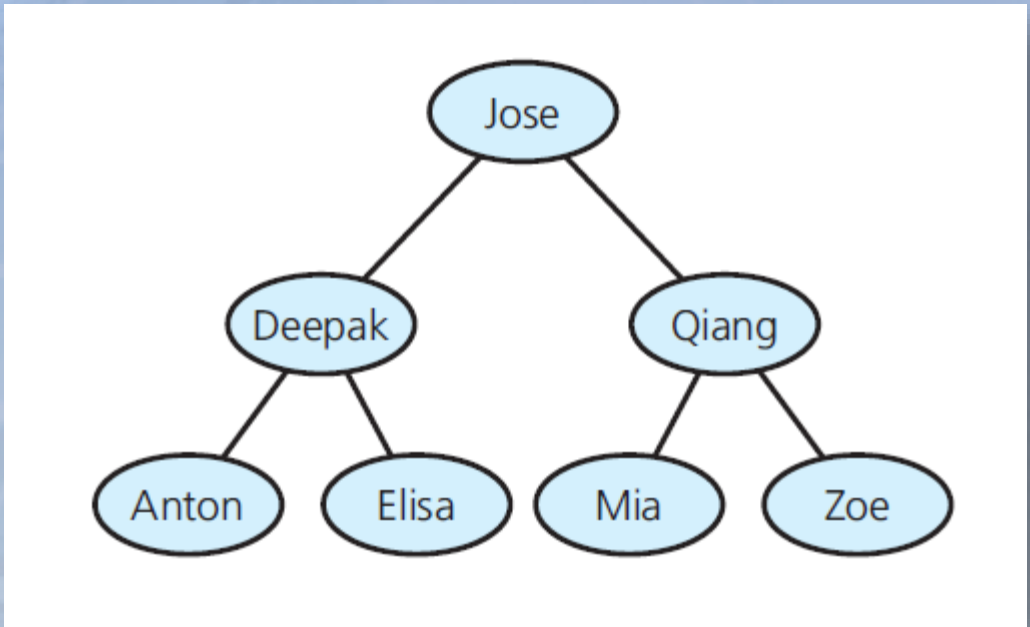
Kinds of Trees



A binary search tree is a binary tree that is, sorted according to the values in its nodes. For each node n, a binary search tree satisfies the following three properties:

1. n ’s value is greater than all values in its left subtree TL
2. n ’s value is less than all values in its right subtree TR
3. Both TL and TR are binary search trees.

FIGURE 15-4 A binary search tree of names

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The Height of Trees

- Level of a node, *n*
  - If *n* is root, level 1
  - If *n* not the root, level is 1 greater than level of its parent
- Height of a tree
  - Number of nodes on longest path from root to a leaf
  - *T* empty, height 0
  - *T* not empty, height equal to max level of nodes

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Recursive definition of height

For binary trees, it is often convenient to use an equivalent recursive definition of height:

- If T is empty, its height is 0.
- If T is a nonempty binary tree, then:
  - height (T) = 1 + max {height (TL), height (TR)}

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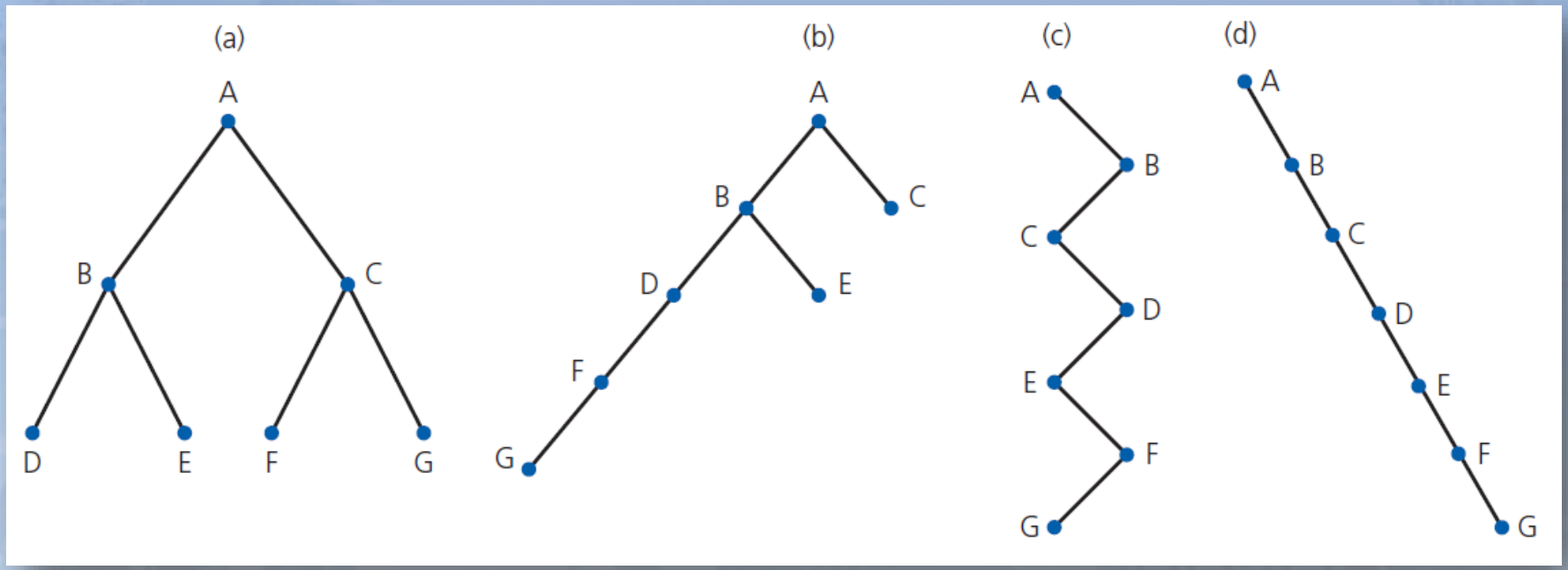
The Height of Trees



The heights are:3, 5, 7, and 7 respectively.

FIGURE 15-5 Binary trees with the same nodes but different heights

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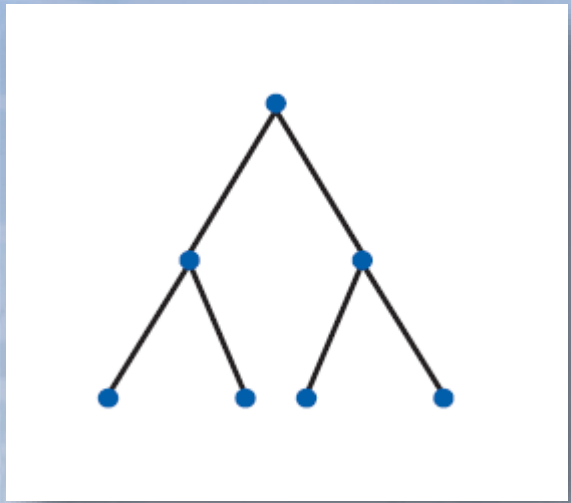
Full, Complete, and Balanced Binary Trees



In a **full** binary tree of height h, all nodes that are at a level less than h have two children each.

FIGURE 15-6 A **full** binary tree of height 3

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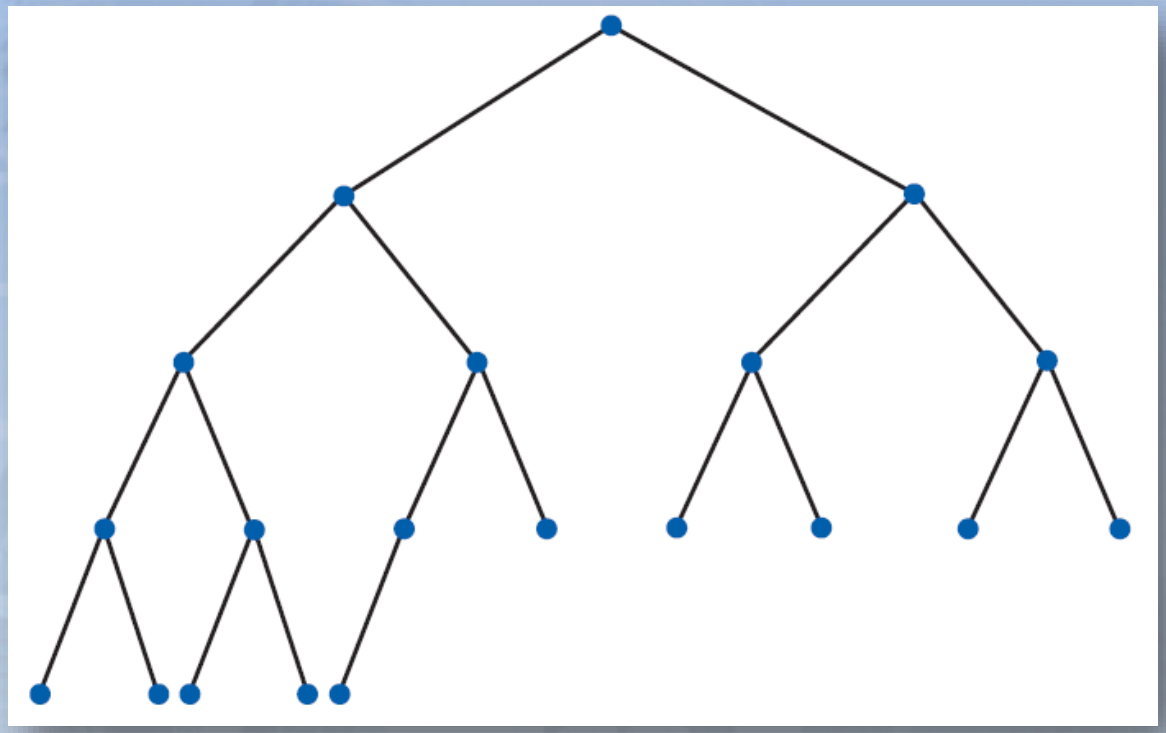
Full, Complete, and Balanced Binary Trees



A **complete** binary tree of height h is a binary tree that is full down to level h – 1, with level h filled in from left to right:

1. All nodes at level h – 2 and above have two children each
2. When a node at level h – 1 has children, all nodes to its left at the same level have two children each
3. When a node at level h – 1 has one child, it is a left child

FIGURE 15-7 A **complete** binary tree

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Full, Complete, and Balanced Binary Trees

- Full binary trees are complete
  - Binary tree of height h with no missing nodes.
  - All leaves are at level h
  - All other nodes each have two children.
- Complete binary trees are balanced
  - Binary tree of height h that is full to level h – 1 and has level h filled in from left to right.
- Balanced binary tree
  - Binary tree in which the left and right subtrees of any node have heights that differ by at most 1

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The Maximum and Minimum Heights of a Binary Tree

- Binary tree with *n* nodes
  - Max height is *n*
- To minimize height of binary tree of *n* nodes
  - Fill each level of tree as completely as possible
  - A complete tree meets this requirement

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The Maximum and Minimum Heights of a Binary Tree

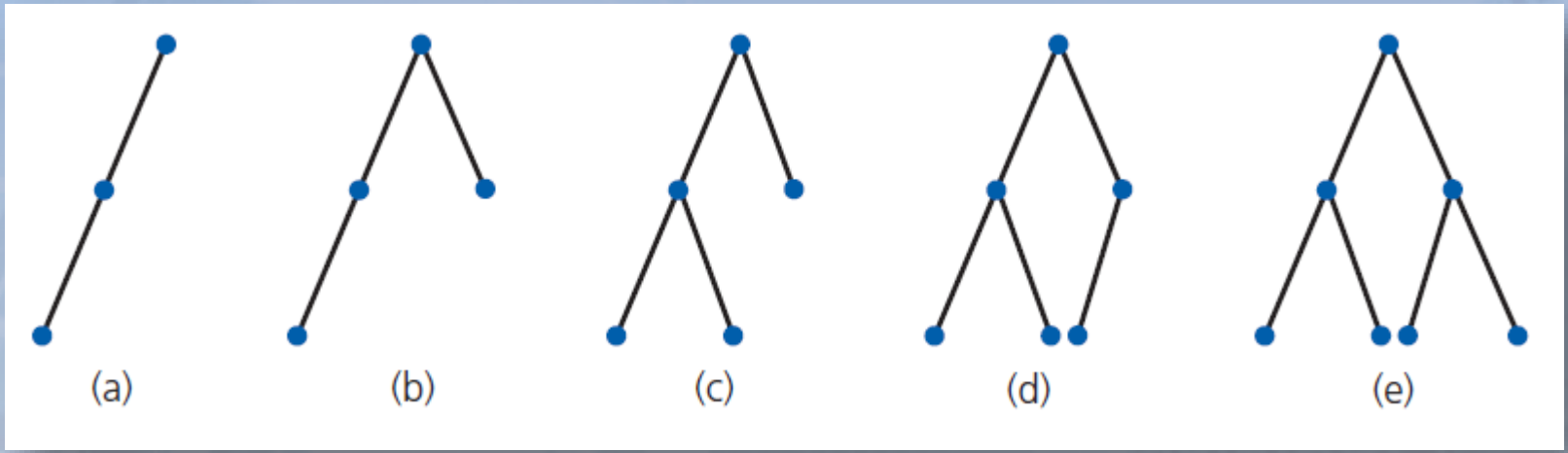

FIGURE 15-8 Binary trees of height 3

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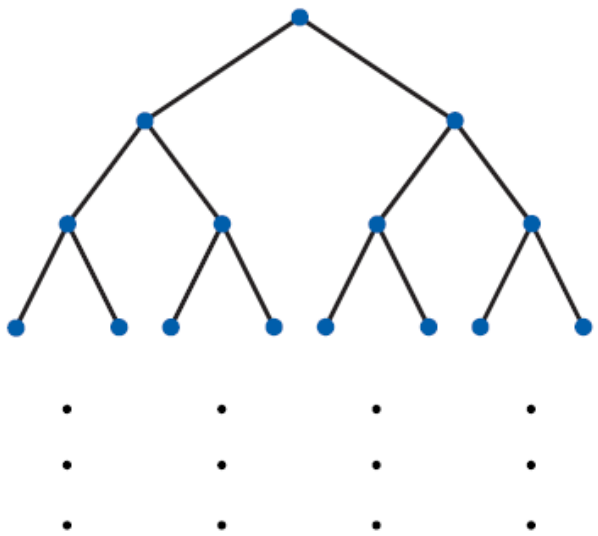
The Maximum and Minimum Heights of a Binary Tree



Level	Number of nodes at this level	Total number of nodes at this level and all previous levels
1	$1 = 2^0$	$1 = 2^1 - 1$
2	$2 = 2^1$	$3 = 2^2 - 1$
3	$4 = 2^2$	$7 = 2^3 - 1$
4	$8 = 2^3$	$15 = 2^4 - 1$
.	.	.
.	.	.
.	.	.
h	2^{h-1}	$2^h - 1$

Maximum # of nodes for a binary tree

FIGURE 15-9 Counting the nodes in a full binary tree of height *h*

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The ADT Binary Tree

- Operations of ADT binary tree
  - Add
  - Remove
  - Set
  - Retrieve data
  - Test for empty
  - Traversal operations that visit every node
- Traversal can visit nodes in several different orders

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Traversals of a Binary Tree

- Options for when to visit the root
  - Preorder: before it traverses both subtrees
  - Inorder: after it traverses left subtree, before it traverses right subtree
  - Postorder: after it traverses both subtrees
- Note traversal is O(*n*)

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Traversals of a Binary Tree

- Pseudocode for general form of a recursive traversal algorithm

```
if (T is not empty)
{
    Display the data in T's root
    Traverse T's left subtree
    Traverse T's right subtree
}
```

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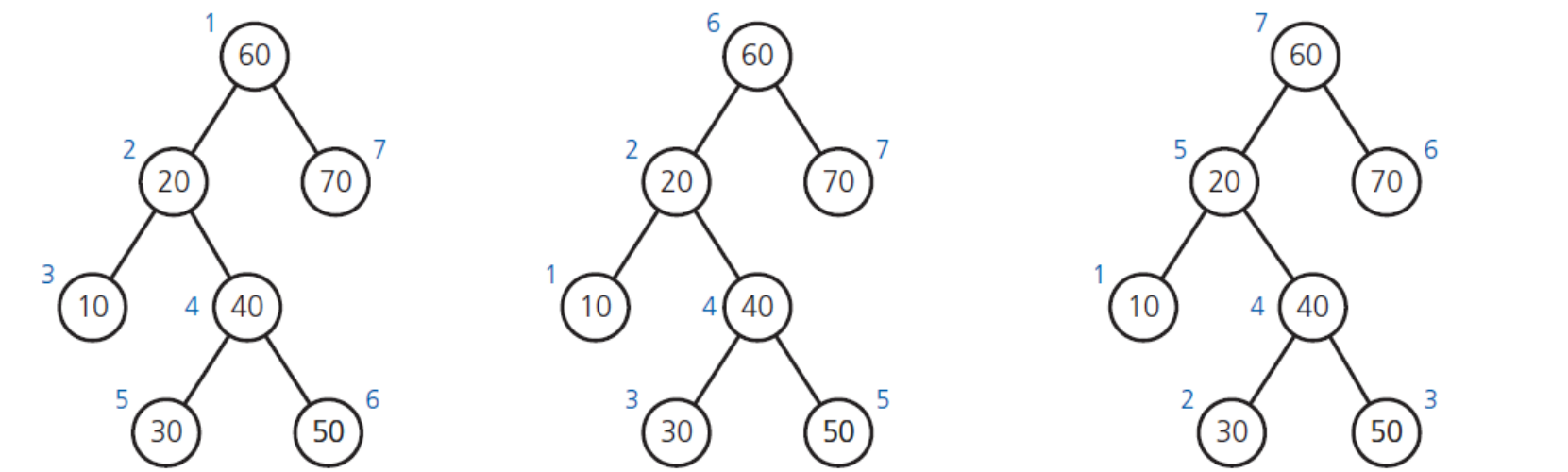
Traversals of a Binary Tree



(a) Preorder: 60, 20, 10, 40, 30, 50, 70 (b) Inorder: 10, 20, 30, 40, 50, 60, 70 (c) Postorder: 10, 30, 50, 40, 20, 70, 60

(Numbers beside nodes indicate traversal order.)

FIGURE 15-11 Three traversals of a binary tree

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Traversals of a Binary Tree

```
// Traverses the given binary tree in preorder.
// Assumes that "visit a node" means to process the node's data item.
preorder(binTree: BinaryTree): void
{
   if (binTree is not empty)
   {
      Visit the root of binTree
      preorder(Left subtree of binTree's root)
      preorder(Right subtree of binTree's root)
   }
}
```

Preorder traversal algorithm

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Traversals of a Binary Tree

```
// Traverses the given binary tree in inorder.
// Assumes that "visit a node" means to process the node's data item.
inorder(binTree: BinaryTree): void
{
   if (binTree is not empty)
   {
      inorder(Left subtree of binTree's root)
      Visit the root of binTree
      inorder(Right subtree of binTree's root)
   }

}
```

Inorder traversal algorithm

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Traversals of a Binary Tree

```
// Traverses the given binary tree in postorder.
// Assumes that "visit a node" means to process the node's data item.
postorder(binTree: BinaryTree): void
{   if (binTree is not empty)
    {
        postorder(Left subtree of binTree's root)
        postorder(Right subtree of binTree's root)
        Visit the root of binTree
    }
}
```

Postorder traversal algorithm

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Binary Tree Operations

```
BinaryTree
---------------------------------------------------------------
+isEmpty(): boolean
+getHeight(): integer
+getNumberOfNodes(): integer
+getRootData(): ItemType
+setRootData(newData: ItemType): void
+add(newData: ItemType): boolean
+remove(target: ItemType): boolean
+clear(): void
+getEntry(target: ItemType): ItemType
+contains(target: ItemType): boolean
+preorderTraverse(visit(item: ItemType): void): void
+inorderTraverse(visit(item: ItemType): void): void
+postorderTraverse(visit(item: ItemType): void): void
```

FIGURE 15-12 UML diagram for the class BinaryTree

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Interface Template for the ADT Binary Tree

```
/** Interface for the ADT binary tree.
 @file BinaryTreeInterface.h */

#ifndef BINARY_TREE_INTERFACE_
#define BINARY_TREE_INTERFACE_
#include "NotFoundException.h"

template<class ItemType>
class BinaryTreeInterface
{
public:
   /** Tests whether this binary tree is empty.
    @return  True if the binary tree is empty, or false if not. */
   virtual bool isEmpty() const = 0;

   /** Gets the height of this binary tree.
    @return  The height of the binary tree. */
   virtual int getHeight() const = 0;

   /** Gets the number of nodes in this binary tree.
    @return  The number of nodes in the binary tree. */
   virtual int getNumberOfNodes() const = 0;
```

LISTING 15-1 An interface template for the ADT binary tree

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Interface Template for the ADT Binary Tree

```
    /** Gets the data that is in the root of this binary tree.
     @pre  The binary tree is not empty.
     @post  The root's data has been returned, and the binary tree is unchanged.
     @return  The data in the root of the binary tree. */
    virtual ItemType getRootData() const = 0;

    /** Replaces the data in the root of this binary tree with the given data,
        if the tree is not empty. However, if the tree is empty, inserts a new
        root node containing the given data into the tree.
     @pre  None.
     @post  The data in the root of the binary tree is as given.
     @param newData  The data for the root. */
    virtual void setRootData(const ItemType& newData) = 0;

    /** Adds the given data to this binary tree.
     @param newData  The data to add to the binary tree.
     @post  The binary tree contains the new data.
     @return  True if the addition is successful, or false if not. */
    virtual bool add(const ItemType& newData) = 0;
```

LISTING 15-1 An interface template for the ADT binary tree

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Interface Template for the ADT Binary Tree

```
    /** Removes the specified data from this binary tree.
     @param target  The data to remove from the binary tree.
     @return  True if the removal is successful, or false if not. */
    virtual bool remove(const ItemType& target) = 0;

    /** Removes all data from this binary tree. */
    virtual void clear() = 0;

    /** Retrieves the specified data from this binary tree.
     @post  The desired data has been returned, and the binary tree
        is unchanged. If no such data was found, an exception is thrown.
     @param target  The data to locate.
     @return  The data in the binary tree that matches the given data.*/
    virtual ItemType getEntry(const ItemType& target) const = 0;
```

LISTING 15-1 An interface template for the ADT binary tree

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Interface Template for the ADT Binary Tree

```
   /** Tests whether the specified data occurs in this binary tree.
    @post  The binary tree is unchanged.
    @param target  The data to find.
    @return  True if data matching the target occurs in the tree, or false if not. */
   virtual bool contains(const ItemType& target) const = 0;

   /** Traverses this binary tree in preorder (inorder, postorder) and
       calls the function visit once for each node.
   @param visit  A client-defined function that performs an operation on
      either each visited node or its data. */
   virtual void preorderTraverse(void visit(ItemType&)) const = 0;
   virtual void inorderTraverse(void visit(ItemType&)) const = 0;
   virtual void postorderTraverse(void visit(ItemType&)) const = 0;

   /** Destroys this tree and frees its assigned memory. */
   virtual ~BinaryTreeInterface() {  }
}; // end BinaryTreeInterface
#endif
```

LISTING 15-1 An interface template for the ADT binary tree

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The ADT Binary Search Tree

- Recursive definition of a binary search tree
  - *n*'s value is greater than all values in its left subtree T_L.
  - *n*'s value is less than all values in its right subtree T_R.
  - Both T_L and T_R are binary search trees.

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The ADT Binary Search Tree

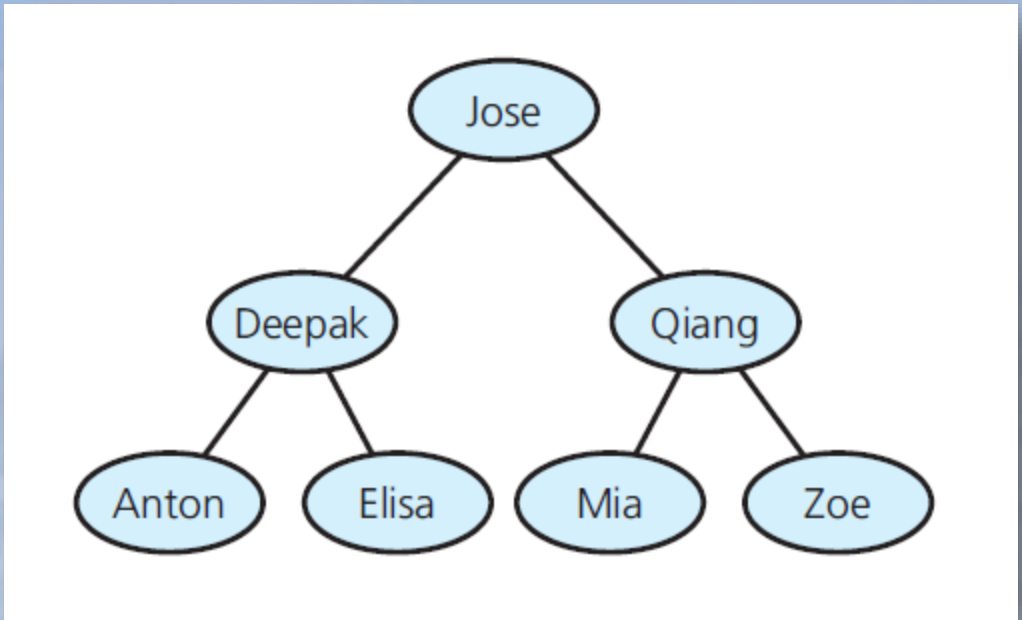

FIGURE 15-13 A binary search tree of names

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Binary Search Tree Operations

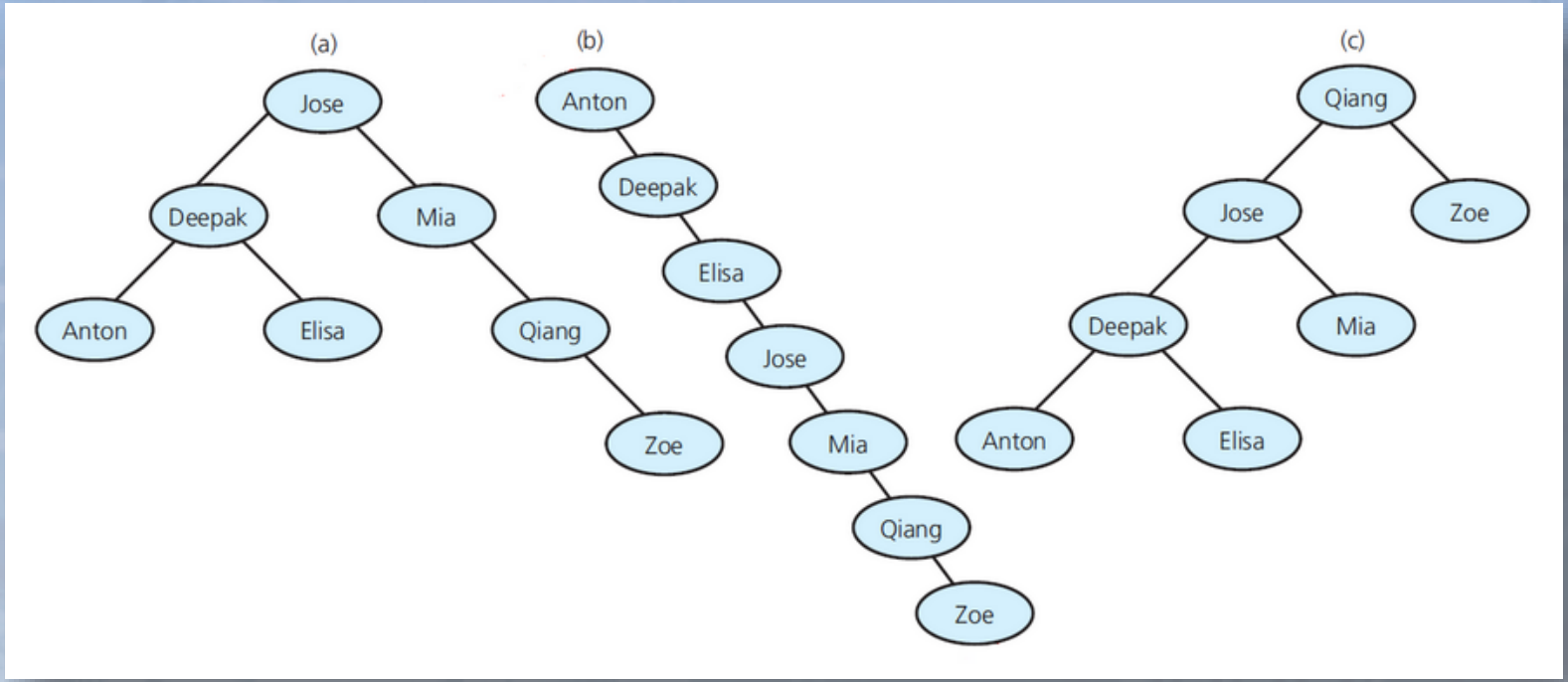

FIGURE 15-14 Binary search trees with the same data as in Figure 15-13

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Binary Search Tree Operations

- Test whether a binary search tree is empty.
- Get the height of a binary search tree.
- Get the number of nodes in a binary search tree.
- Get the data in a binary search tree's root.
- Add the given data item to a binary search tree.
- Remove the specified data item from a binary search tree.
- Remove all data items from a binary search tree.
- Retrieve the specified data item in a binary search tree.
- Test whether a binary search tree contains specific data.
- Traverse the nodes in a binary search tree in preorder, inorder, or postorder.

Operations that define the ADT binary search tree

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Searching a Binary Search Tree

Search algorithm for a binary search tree

```
// Searches the binary search tree for a given target value.
search(bstTree: BinarySearchTree, target: ItemType)
{
   if (bstTree is empty)
      The desired item is not found
   else if (target == data item in the root of bstTree)
      The desired item is found
   else if (target < data item in the root of bstTree)
      search(Left subtree of bstTree, target)
   else
      search(Right subtree of bstTree, target)
}
```

- This algorithm will work regardless of the shape of a binary search tree.
- The worse case of a binary search tree is O(n), when it is not balanced.
- The average time complexity for a binary search tree is O(log n)
- The best case of a binary search tree is O(1).

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Binary Search Tree

https://en.wikipedia.org/wiki/Binary_search_tree

Binary search tree		
Type	tree	
Invented	1960	
Invented by	P.F. Windley, A.D. Booth, A.J.T. Colin, and T.N. Hibbard	
Time complexity in big O notation		
Operation	Average	Worst case
Search	$O(\log n)$	$O(n)$
Insert	$O(\log n)$	$O(n)$
Delete	$O(\log n)$	$O(n)$
Space complexity		
Space	$O(n)$	$O(n)$

- The AVL tree was invented by Georgy Adelson-Velsky and Evgenii Landis in 1962 for the efficient organization of information. It was the first self-balancing binary search tree to be invented.
- Various height-balanced binary search trees were introduced to confine the tree height, such as AVL trees (1962), Treaps (1989) (https://en.wikipedia.org/wiki/Treap), and red–black trees (1978) (https://en.wikipedia.org/wiki/Red%E2%80%93black_tree).

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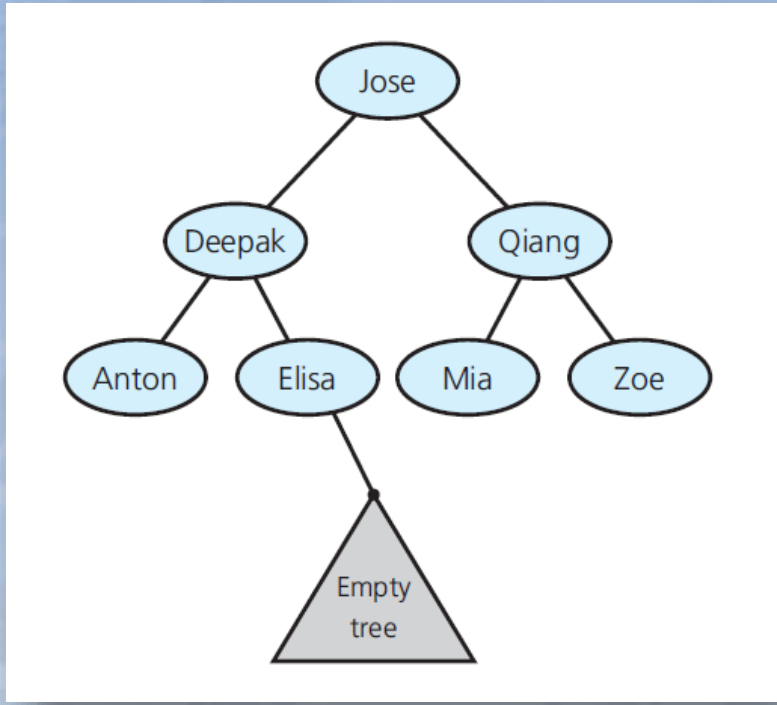
Creating a Binary Search Tree



nameTree = a new, empty binary search tree

nameTree .insert(“Jose")

nameTree .insert(“Deepak")

nameTree .insert(“Anton")

nameTree .insert("Elisa")

nameTree .insert(“Qiang")

nameTree .insert(“Mia")

nameTree .insert(“Zoe")

nameTree .insert(“Finn")

FIGURE 15-16 Empty subtree where the search algorithm terminates when looking for Finn

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Traversals of a Binary Search Tree

```
// Traverses the given binary tree in inorder.
// Assumes that "visit a node" means to process the node's data item.
inorder(binTree: BinaryTree): void
{
   if (binTree is not empty)
   {
      inorder(Left subtree of binTree's root)
      Visit the root of binTree
      inorder(Right subtree of binTree's root)
   }

}
```

Inorder traversal of a binary search tree visits tree's nodes in sorted search-key order

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Efficiency of Binary Search Tree Operations

- Max number of comparisons for retrieval, addition, or removal
  - The height of the tree
- Adding entries in sorted order
  - Produces maximum-height binary search tree
- Adding entries in random order
  - Produces near-minimum-height binary search tree

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Efficiency of Binary Search Tree Operations

Operation	Average case	Worst case
Retrieval	O(log *n*)	O(*n*)
Addition	O(log *n*)	O(*n*)
Removal	O(log *n*)	O(*n*)
Traversal	O(*n*)	O(*n*)

FIGURE 15-17 The Big O for the retrieval, addition, removal, and traversal operations of the ADT binary search tree

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End

Chapter 15

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